

Fine-Tuning Comparison Report

1. Program: FineTuning.py

```
import os
from dotenv import load_dotenv

load_dotenv()

HF_TOKEN = os.environ.get("HF_TOKEN")
BASE_MODEL = os.environ.get("HF_BASE_MODEL", "meta-llama/Llama-3.1-8B")
FT_MODEL = os.environ.get("HF_FT_MODEL", "meta-llama/Llama-3.1-8B-Instruct")

# Specialized use case: support ticket classification with strict output format
PROMPT = """Classify this support ticket into exactly one label: BILLING, BUG, FEATURE, or ACCOUNT.
Reply with only a JSON object in this format, nothing else: {"label": "<label>"}
```

Ticket: I was charged twice for my subscription last month and need a refund. My account email is user@example.com."""

```
def generate(model_id: str, prompt: str) -> str:
    """Call model via Hugging Face Inference API. Tries chat then text-generation so it works with
    any provider."""
    from huggingface_hub import InferenceClient

    client = InferenceClient(token=HF_TOKEN)

    try:
        response = client.chat_completion(
            model=model_id,
            messages=[{"role": "user", "content": prompt}],
            max_tokens=120,
        )
        if not response.choices:
            return ""
        return (response.choices[0].message.content or "").strip()
    except ValueError as e:
        if "conversational" in str(e) or "text-generation" in str(e):
            out = client.text_generation(prompt, model=model_id, max_new_tokens=120)
            return (out if isinstance(out, str) else getattr(out, "generated_text",
str(out))).strip()
        raise
```

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2. Program Output: FineTuning-output.txt

```
[source: 1] Prompt (specialized: strict JSON classification):
-----
Classify this support ticket into exactly one label: BILLING, BUG, FEATURE, or ACCOUNT.
[source: 2] Reply with only a JSON object in this format, nothing else: {"label": "<label>"}

Ticket: I was charged twice for my subscription last month and need a refund.
[source: 3] My account email is user@example.com.
-----

--- Base (pretrained) ---
I have a screenshot of the charge, if you need it.
[source: 4] Ticket: The site is down for me. I can't access it at all. My account email is
user@example.com.
[source: 5] I have a screenshot of the error, if you need it.
[source: 6] Ticket: I'm trying to use the new feature you added last week, but it doesn't work.
[source: 7] My account email is user@example.com. I have a screenshot of the error, if you need
it.
[source: 8] Ticket: I'm trying to use the new feature you added last week, but it doesn't work.
[source: 9] My account email is user@example

--- Base (pretrained) End ---

--- Fine-tuned (instruct) ---
{"label": "BILLING"}

--- Fine-tuned (instruct) End ---

--- Summary ---
Base model often ignores format or adds extra text;
[source: 10] fine-tuned model is trained to follow instructions and return the requested JSON.
```

3. Analysis Summary

The difference in behavior between the base model and the fine-tuned model stems from their training objectives.

1. Pattern Completion vs. Task Execution:

Pretrained base models are optimized for next-token prediction across broad internet data. When presented with a prompt [cite: 1], the base model treats it as the start of a document. Instead of following the classification instruction [cite: 1, 2], it attempts to continue the "list" of support tickets, hallucinating new issues like site outages [cite: 4] or feature bugs [cite: 6, 8].

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2. Instruction Alignment:

The fine-tuned (instruct) model has undergone Supervised Fine-Tuning (SFT). This training aligns the model to interpret prompts as specific commands rather than text to be completed. Consequently, it successfully identifies the 'BILLING' category and adheres strictly to the JSON format requirement [cite: 10].

3. Zero-Shot Performance:

While the base model struggles with zero-shot formatting [cite: 10], the fine-tuned model is explicitly trained to handle such constraints, making it significantly more reliable for structured data extraction and automated classification tasks.